

computer_proj4

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```
options(digits=10)
```

```
# Define function to complete importance sampling for the integral to be estimated
importance_sample <- function(n) {
  # Sample uniform iids
  u <- runif(n)

  # Compute x rvs
  x <- sin(pi*(u-0.5))

  # Compute estimate for I
  I <- 1/n * sum(cosh(x))

  return(I)
}
```

```
# Compute actual (according to R integrate() function) value of I to compare with estimate
f <- function(x) cosh(x)/sqrt(1-x^2)
actual <- (1/pi)*integrate(f, lower=-1, upper=1)$value
print(actual)
```

```
## [1] 1.266065878
```

```
n_vals <- c(10, 100, 1000, 10000, 100000, 1000000000)

for(n in n_vals) {
  # Compute I for given n
  I <- importance_sample(n)

  # Compute absolute and relative error
  abs_err <- abs(actual-I)
  rel_err <- abs_err/actual

  print(paste("n =", n, "estimate =", I, "absolute error =", abs_err, "relative error = ", rel_e
rr))
  print("-----")
  print("-----")
}
```

```
## [1] "n = 10 estimate = 1.3469506083407 absolute error = 0.0808847307828149 relative error =
0.063886668313685"
## [1] "-----"
## [1] "n = 100 estimate = 1.30040215002672 absolute error = 0.0343362724688372 relative error =
0.0271204469510454"
## [1] "-----"
## [1] "n = 1000 estimate = 1.26029035335424 absolute error = 0.00577552420363636 relative error
= 0.00456178806017329"
## [1] "-----"
## [1] "n = 10000 estimate = 1.26613633934565 absolute error = 7.04617877667069e-05 relative err
or = 5.56541243356316e-05"
## [1] "-----"
## [1] "n = 1e+05 estimate = 1.26583467259243 absolute error = 0.000231204965454479 relative err
or = 0.000182616852371419"
## [1] "-----"
## [1] "n = 1e+09 estimate = 1.26605141718671 absolute error = 1.44603711753621e-05 relative err
or = 1.14214998063567e-05"
## [1] "-----"
```

Remarks

As we can see, sampling with even only $n = 10$ points only results in a relative error on the order of 10% but this error grows smaller and smaller as we increase n towards ∞ , coming within a small fraction of a percent of the true value of the integral calculated by R.